

KISHORE KUMAR RAMKUMAR

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EDUCATION

Indiana University Bloomington, Indiana

Master of Science, Data Science ; GPA - 3.5

Aug 2024 - May 2026

Indiana

Sri Ramachandra Institute of Higher Education and Research, Chennai

Bachelor of Technology, Computer Science Specializing in AI & ML

Jul 2021 - Jun 2024

Chennai

TECHNICAL SKILLS

- **Programming Languages & Development:** Python, Java, R, SQL, Postgres, MySQL, JavaScript, ReactJS, Git
- **Software Engineering Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming(OOP), Multithreading & Concurrency, System Design, Debugging, Unit Testing
- **Backend & Distributed Systems:** REST APIs, Microservices, Distributed Systems, Event-Driven Architecture(s)
- **Cloud & DevOps:** AWS, EC2, Lambda, S3, EKS, SQS, DynamoDB, CloudWatch, Docker, Kubernetes, CI/CD, Linux
- **Data & Streaming Systems:** Kafka, Apache Spark, ETL Pipelines, Parallel Computing, Ray
- **ML Systems (Supporting):** PyTorch, TensorFlow, RAG, FAISS, Model Serving, NLP (spaCy, NLTK)

EXPERIENCE

Media School, Communications, Indiana University

Research Assistant (Applied ML Systems)

Jan 2026 - Present

Bloomington, IN

- Built **automated NLP data pipelines** in **Python** (speech-to-text, preprocessing, feature extraction) to convert large-scale TV audio into **model-ready structured datasets**, eliminating manual annotation workflows.
- Designed **scalable preprocessing and EDA pipelines** using **Pandas, NumPy, spaCy/NLTK** and **R**, standardizing **10+ years** of text data for consistent downstream consumption.
- Extended longitudinal datasets beyond 2015 by engineering **metadata extraction pipelines (5K+ records)**, improving data coverage and analytical reliability through **data-driven automation**.

Indiana University O'Neill School of Public & Environmental Affairs

Research Assistant (Data Engineer)

May 2025 - Aug 2025

Bloomington, IN

- Built a **distributed data processing pipeline** in **Python**, processing **530+ GB** of CMIP6 precipitation data and publishing **versioned NetCDF outputs** for cross-system analysis.
- Improved pipeline performance by **32%** by **parallelizing workloads** across **HPC nodes** and optimizing **I/O throughput** on Slate storage.
- Implemented **time-series aggregation and data validation pipelines**, generating 3/6/12-month rolling metrics and ensuring reliability of downstream analytics.

MRAR Intellect Solutions

Machine Learning Associate Engineer Intern

Jan 2024 - Aug 2024

Mississauga, Canada

- Developed a car-manual RAG backend using **FastAPI, FAISS-GPU**, and **PyTorch** on **AWS**, enabling low-latency semantic retrieval and improved response accuracy.
- Deployed **TorchServe + FastAPI** services on **AWS EKS** behind **ALB** with autoscaling and **CloudWatch** monitoring, maintaining **95%** of GPU inferences under **150 ms** using CUDA half-precision.
- Built **CI/CD pipelines** with **Docker** and **ECR**, increasing GPU evaluation throughput by **18%** via dataset sharding and batch optimization.

PROJECTS

TruthLens | <https://github.com/kishuxz/aws-hackathon-truthlens>

- Led development of a **cloud-native reliability service** on **AWS Bedrock**, generating auditable **0–100 scores** with citations and consistency checks.
- Built a **React frontend** and **serverless REST API** using **AWS Lambda** and **S3**, implementing **IAM least privilege**, validation, and monitoring to meet latency/SLO targets.
- Designed an **end-to-end data processing pipeline** integrating **SageMaker OCR** and policy-driven scoring, enabling reliable and traceable outputs.

Cloud Job Scheduler (AWS) | <https://github.com/kishuxz/Job-Scheduler>

- Built a **cloud-native distributed job scheduling service** using **AWS Lambda, SQS, DynamoDB**, and **ECS**.
- Implemented **idempotent job submission** to safely handle retries and prevent duplicate execution in a distributed system.
- Designed **fault-tolerant job execution** with asynchronous workers and explicit state transitions (**PENDING, RUNNING, SUCCEEDED, FAILED**).
- Owned **deployment, monitoring, and troubleshooting** using **CloudWatch logs and metrics**, ensuring service reliability.

Cloud Based Smart Energy Management System | <https://github.com/kishuxz/smart-energy-management-system>

- Built an **event-driven data pipeline** using **Kafka** and **Spark**, processing **10,000+ time-series energy records** across hourly, daily, and rolling windows.
- Implemented **distributed aggregation and feature pipelines** in **Spark**, enabling scalable analytics and baseline demand forecasting.
- Exposed processed outputs for downstream consumption and visualization, supporting operational analysis and planning workflows

CERTIFICATIONS

- **AWS Certified : Machine Learning Engineer - Associate: November 2025 - November 2028**
- **AWS Certified : Developer - Associate: January 2026 - January 2029**